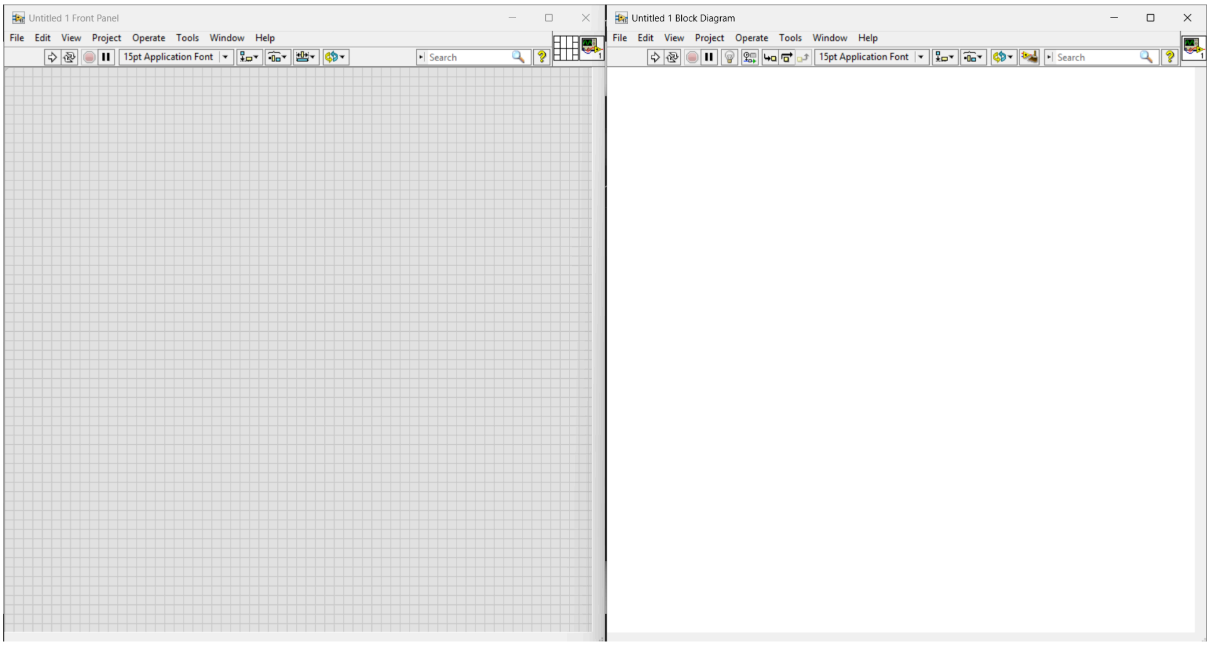
# LabVIEW Assignment 4

## 1)Create a Simulation for Temperature Conversion.

### 1.1) Convert Fahrenheit to Celsius

STEPS

1. Create new **blank VI** (Virtual Instrument) **project**. It will open two windows: Front Panel & Block Diagram. The User Interface is made on the Front Panel and the connections are made on the Block Diagram.
2. Use **Ctrl+t** shortcut of LabVIEW platform to split view both the Panes (Front Panel & Block Diagram)



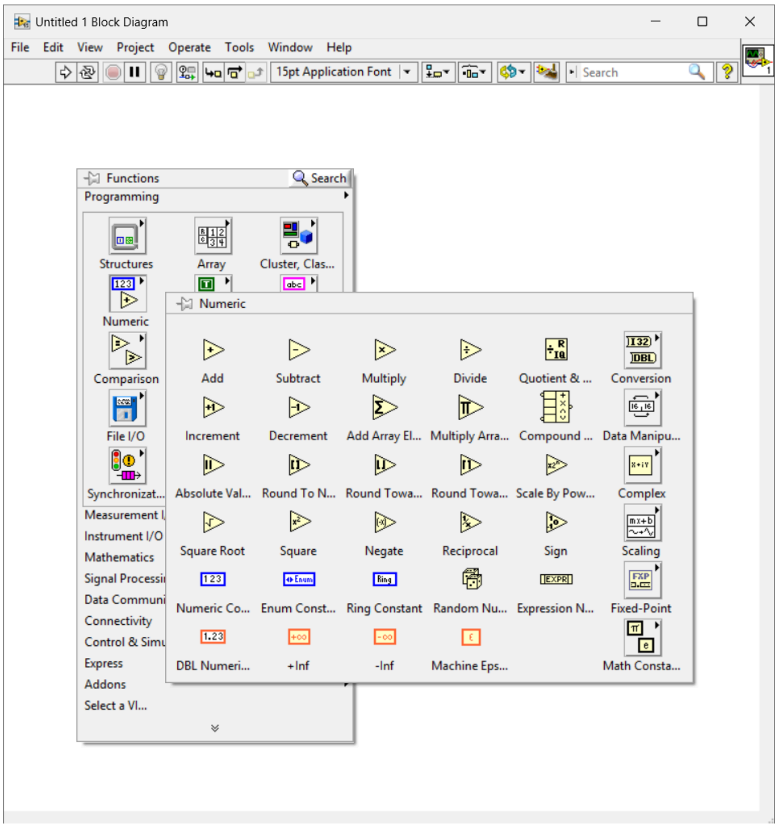

3. Write down the **required formula** on a rough sheet. Make a note of operators/logic to be used in the simulation (Addition, Subtraction, Multiplication, Division, Quotient, Remainder)

$$C = \frac{5}{9} * (F - 32)$$

In this case Division, Multiplication and Subtraction operator is used.

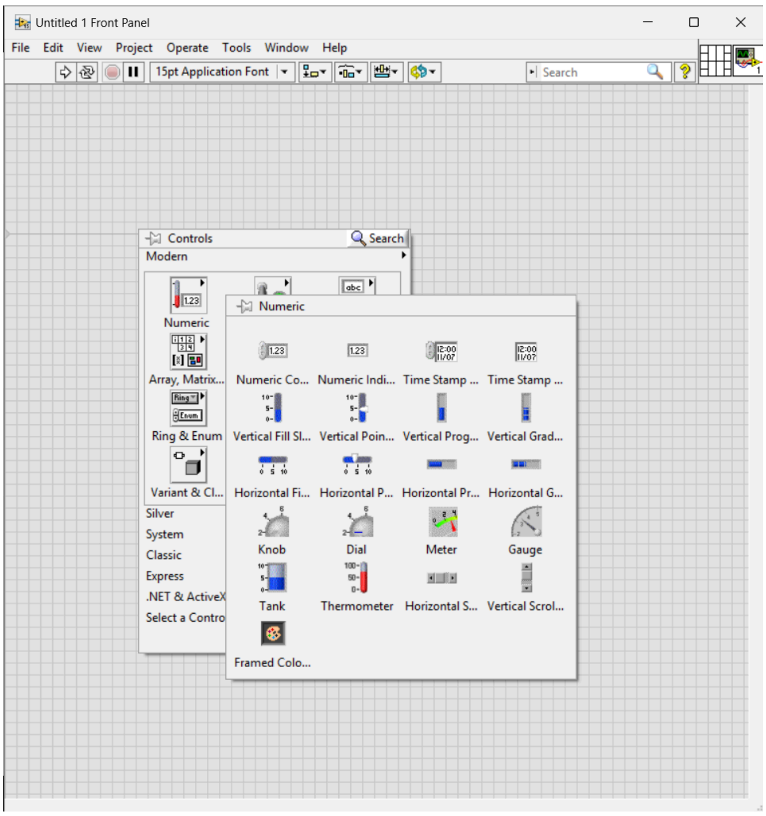
4. Add the **mathematical operators and constants** (5, 9, 32) from the right-click menu (Functions Menu) of the Block-Diagram Panel one-by-one.

***Right-Click Functions menu of Block Diagram > Programming > Numeric > (Division, Multiplication, Subtraction, Numeric Constants)***




## 5. Controls Interface for Users

To input temperature in Fahrenheit & output temperature in Celsius, **Controls** have to added. And to show the corresponding numeric value **Numeric Indicators** have to added.



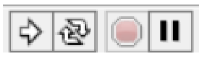
***Right Click Control Menu > Modern > Numeric > (Vertical Fill Slide, Thermometer, Numeric Indicator)***

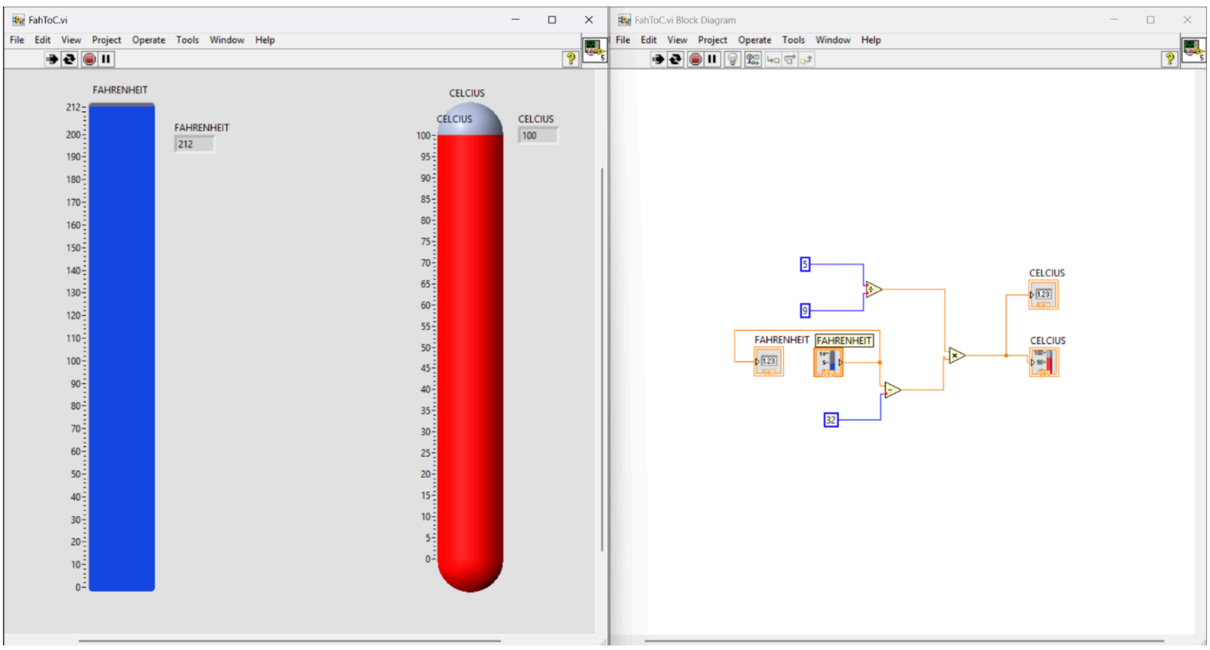
Upon adding the Controls their corresponding block diagram also appear on the Block Diagram Panel.

6. To make connections use **Thread** from the **Shift+Right-Click Menu** on the Block Diagram Panel 

   The colour of thread reveals the Data Type carried by it.

   BLUE ---- integer
   ORANGE --- float
   GREEN --- boolean
   PINK --- string

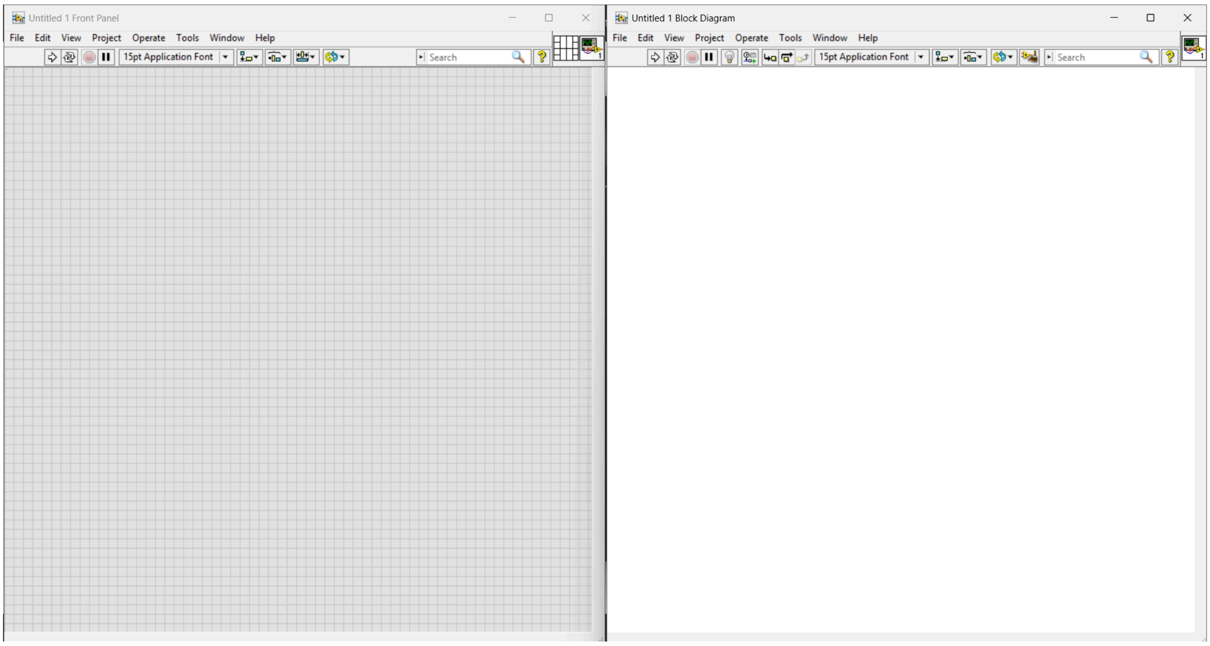
7. To run the simulation, use the **Run Continuously** option. 
   - *Use the **slider** to input fahrenheit temperature and **Thermometer** to view output celcius temperature (ON FRONT PANEL)*



## 1.2) Convert Celsius to Fahrenheit

STEPS

1. Create new **blank VI (**Virtual Instrument**) project**. It will open two windows: Front Panel & Block Diagram. The User Interface is made on the Front Panel and the connections are made on the Block Diagram.
2. Use **Ctrl+t** shortcut of LabVIEW platform to split view both the Panes (Front Panel & Block Diagram)



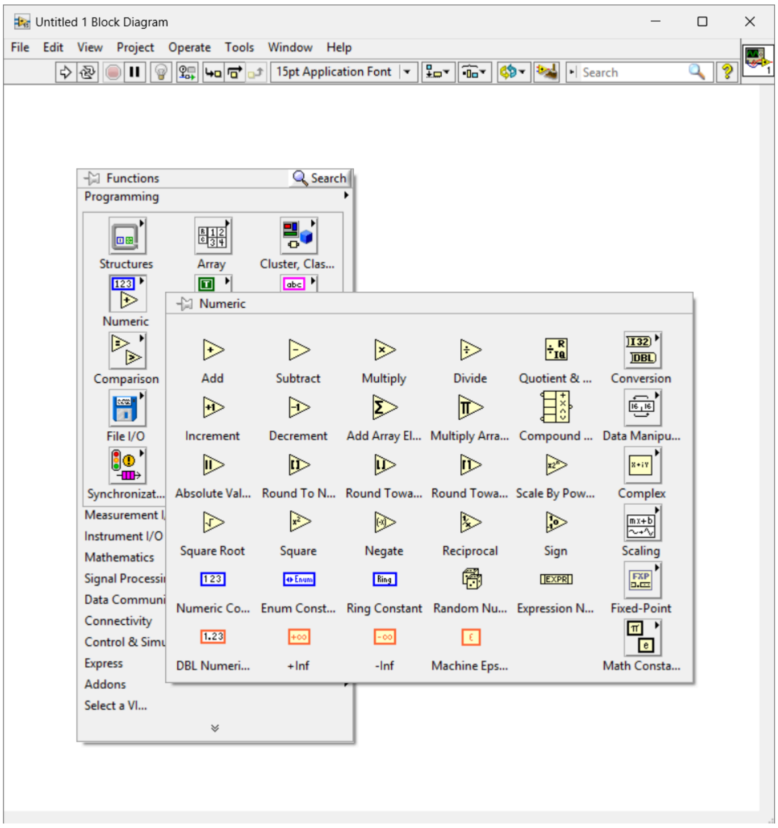

3. Write down the **required formula** on a rough sheet. Make a note of operators/logic to be used in the simulation (Addition, Subtraction, Multiplication, Division, Quotient, Remainder)

$$F = \frac{9}{5} * C + 32$$

   In this case Division, Multiplication and Subtraction operator is used.

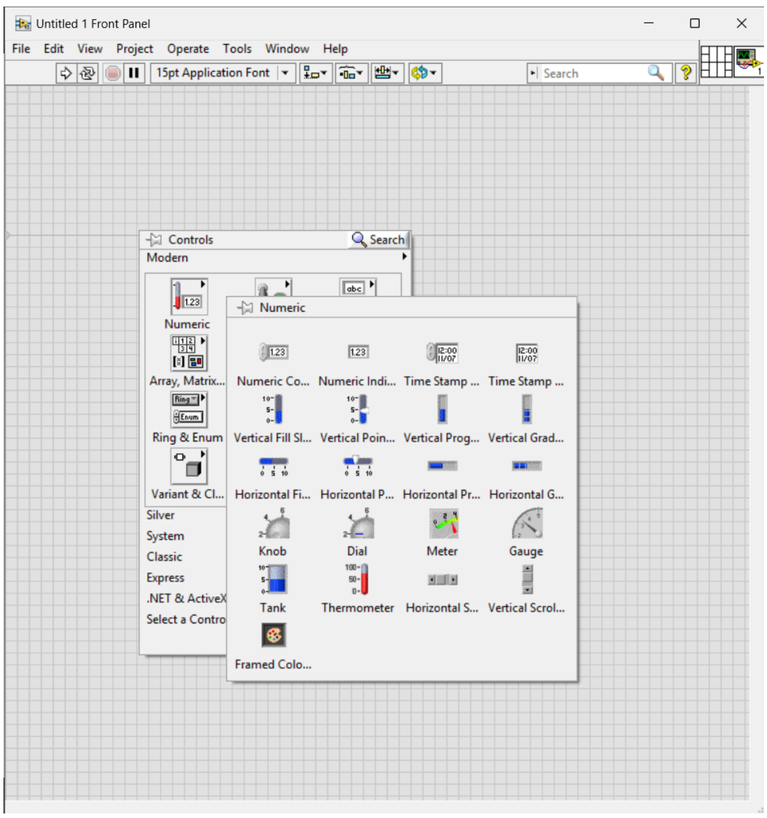
4. Add the **mathematical operators and constants** (9, 5, 32) from the right-click menu (Functions Menu) of the Block-Diagram Panel one-by-one.

   ***Right-Click Functions menu of Block Diagram > Programming > Numeric > (Division, Multiplication, Addition, Numeric Constants)***




## 5. Controls Interface for Users

To input temperature in Celsius & output temperature in Fahrenheit, **Controls** have to added. And to show the corresponding numeric value **Numeric Indicators** have to added.



***Right Click Control Menu > Modern > Numeric > (Vertical Fill Slide, Thermometer, Numeric Indicator)***
Upon adding the Controls their corresponding block diagram also appear on the Block Diagram Panel.

6. To make connections use **Thread** from the **Shift+Right-Click Menu** on the Block Diagram Panel 

    - The colour of thread reveals the Data Type carried by it.

      BLUE ---- integer
      ORANGE --- float
      GREEN --- boolean
      PINK --- string

7. To run the simulation, use the **Run Continuously** option. 

    - *Use the **slider** to input Celsius temperature and **Thermometer** to view output Fahreheit temperature (ON FRONT PANEL)*